

Smart Parking for Medical Cities via Passenger-Aware Allocation and Fast Grouping-Permutation Heuristics

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ABSTRACT Medical cities serve large populations and experience high parking demand due to heterogeneous visitor profiles and time constraints. This paper addresses the parking allocation problem in such environments by proposing a grouping-based optimization framework that allocates spaces according to the total number of passengers per vehicle rather than vehicle count alone. The proposed system integrates passenger-count information and real-time occupancy data through a coordinated architecture involving booking interfaces, IoT-based sensing, and a centralized allocation engine. Six novel algorithms are developed to achieve equitable load distribution across parking zones. Their performance is evaluated using nine instance classes and 2,790 test cases. Experimental results demonstrate that the *GVF* algorithm achieves the highest best-solution percentage (68.2%) and the lowest average gap among the proposed algorithms, confirming its scalability and robustness in large-scale medical city environments.

INDEX TERMS Parking management, medical cities, Internet of Things (IoT), parking allocation, vehicle occupancy, load balancing, smart healthcare systems, grouping-based optimization.

I. INTRODUCTION

Medical cities represent large-scale, integrated healthcare ecosystems designed to provide specialized medical, surgical, and emergency services across multiple disciplines [1]. These facilities typically comprise major hospitals, specialized treatment centers, research institutes, and academic units that collectively support advanced clinical care and medical innovation. Due to their comprehensive service offerings, medical cities attract substantial daily visitor volumes, including patients, healthcare professionals, and accompanying family members.

A significant proportion of visitors arrive by private vehicle, creating high parking demand concentrated within limited time windows. Scheduled appointments, emergency visits, and access to specialized clinics generate dynamic and heterogeneous traffic patterns. Consequently, efficient parking management is a critical operational challenge that affects

service accessibility, patient satisfaction, and overall system efficiency [2], [3].

Traditional parking systems allocate spaces based solely on vehicle count, disregarding passenger occupancy and destination-related characteristics. In medical environments, this approach may lead to inefficient space utilization, congestion near critical service areas, increased search time, and delayed access to care. These challenges are further exacerbated by fluctuating demand, heterogeneous patient conditions, and spatial constraints within large healthcare campuses [4].

This paper addresses the parking allocation problem in medical cities by introducing a grouping-based optimization framework that assigns parking spaces according to the total number of passengers per vehicle [5]. By shifting from vehicle-based to passenger-based allocation, the proposed approach seeks to improve load balancing, facilitate access to

critical service zones, and maximize the effective use of parking space.

Such an allocation process requires the management of dynamic data streams, the integration of reservation systems with real-time sensing technologies, and the equitable distribution of parking resources under capacity constraints. To address these operational complexities, the proposed system incorporates intelligent scheduling mechanisms supported by IoT-enabled sensing and algorithmic optimization.

The primary contributions of this study are summarized as follows:

- *Passenger-Based Allocation Framework*: A novel parking management model that prioritizes the total passenger count rather than vehicle count alone, thereby improving space utilization and promoting equitable distribution.
- *Integrated Intelligent Architecture*: A coordinated system architecture that combines booking interfaces, real-time IoT sensing, validation mechanisms, and a centralized allocation engine for dynamic parking assignment.
- *Load Balancing via a Grouping Strategy*: A grouping-based optimization methodology designed to minimize imbalance across parking zones by accounting for vehicle occupancy characteristics.
- *Four-Stage Grouping Procedure*: A structured workflow involving prioritization, dataset segmentation, group re-ordering, and scheduling evaluation to explore multiple allocation configurations.
- *Six Grouping–Permutation Algorithms*: A family of six algorithmic variants derived through the systematic combination of grouping orders and sorting strategies to comprehensively explore the solution space.
- *Competitive Performance Against the Baseline*: Across 2,790 benchmark instances, GV_b (defined in Section VI-I-D) matched or outperformed the baseline MR [6] in 1,547 instances (55.4%). Specifically, GV_b outperformed MR in 57 instances (2.0%) and matched it in 1,490 instances (53.4%), whereas MR performed better in 1,243 instances (44.6%). In addition, GV_b required substantially less execution time, supporting its suitability for real-time parking allocation.

The remainder of this paper is organized as follows. Section II reviews the related work and identifies the research gap. Section III formulates the parking allocation problem. Section IV presents the proposed system architecture, including component interactions, data flow, and operational adaptability. Section V introduces the grouping routine. Section VI presents the six proposed algorithms for the parking allocation problem. Section VII reports the experimental results and comparative analysis. Finally, Section VIII concludes the paper and outlines future research directions.

II. RELATED WORK

Early research on smart parking systems focused on integrating sensing technologies, wireless communication, and centralized control mechanisms to reduce congestion and

search time in urban environments. One of the earliest smart parking infrastructures was presented in [7], proposing a sensor-based detection system combined with centralized allocation and reservation mechanisms. Such foundational frameworks established the architectural backbone of modern smart parking systems by emphasizing real-time occupancy detection, reservation management, and decision-support platforms.

Subsequent developments incorporated Internet of Things (IoT) paradigms and dynamic load-balancing scheduling protocols to improve scalability and interoperability in distributed environments [8], [9]. Comprehensive surveys have synthesized advances in architectural design, sensing technologies, and deployment challenges, while highlighting persistent research gaps related to scalability, robustness, and real-world implementation constraints [10]. These studies provide important historical and theoretical grounding for contemporary optimization-driven allocation strategies.

Specialized high-demand environments, such as medical cities, offer a compelling context for the application and evaluation of grouping-based optimization strategies aimed at improving parking management.

To address high vehicle turnover and critical accessibility requirements, medical centers can adopt IoT-based smart parking systems to maximize space efficiency and reduce congestion. For example, IoT sensors that provide real-time parking availability data have been used to improve traffic flow and reduce vehicle idling time [11]. Performance-based parking frameworks that emphasize adaptability and cost-effectiveness have also been proposed to enhance accessibility and sustainability [12].

Optimization-based approaches have gained significant attention. Hybrid machine learning and optimization models, such as Genetic Algorithm–Radial Basis Function (GA-RBF) models and swarm intelligence algorithms for dynamic parking lot scheduling, have been proposed to minimize fuel consumption, optimize space utilization, and reduce travel time [13], [14]. Reinforcement learning approaches have further introduced adaptive allocation mechanisms for dynamic environments [15].

Recent work has also explored constraint optimization formulations for dynamic parking allocation by modeling the problem under real-time system and user constraints to improve efficiency [16]. Although these models improve allocation performance, they generally focus on generic urban settings and do not incorporate domain-specific operational constraints such as those encountered in medical cities.

Although parking allocation has been studied extensively, many existing works emphasize reservation efficiency and capacity utilization. For instance, Zulfiqar et al. [17] discussed reservation mechanisms based on driver requirements, while Abdeen et al. [18] proposed a hierarchical multi-criteria decision model using historical traffic data. Dynamic pricing mechanisms have also been investigated to regulate demand during peak hours [19]. In addition, IoT- and IIoT-based

systems have been employed to enhance parking allocation through real-time monitoring [15], [20].

Other studies have proposed real-time smart parking control and monitoring architectures that integrate IoT sensors, cloud platforms, and adaptive guidance systems to provide live occupancy updates and responsive management capabilities [21]. While such systems improve monitoring and operational visibility, allocation optimization is often treated as a secondary component rather than as a structured multi-constraint decision problem.

The allocation of parking spaces in medical facilities introduces additional operational constraints, including appointment schedules, patient conditions, and clinic destinations. In [22], cumulative prospect theory was employed to model hospital parking allocation under time-window constraints with profit-maximization objectives. Although this approach is tailored to medical facilities, its focus remains primarily on utilization efficiency and revenue generation rather than on equitable accessibility.

Equitable distribution and load-balancing concepts have also been investigated across multiple domains [23], [24], [25]. Drawing inspiration from these advances, researchers have sought to apply analogous techniques to mitigate vehicular congestion and prevent the overloading of local facilities. Specifically, in the parking context, dynamic allocation algorithms have been proposed to balance occupancy across facilities while maximizing utilization efficiency [6], [26]. However, these methods do not explicitly incorporate multi-attribute grouping criteria such as passenger count, clinic destination, appointment timing, and patient status.

Recent advances in Intelligent Transportation Systems (ITS) have integrated machine learning, edge computing, and game-theoretic routing into traffic optimization frameworks [27], [28]. Reinforcement learning and aggregation-based traffic control strategies have also been introduced [29], [30]. UAV-based AI traffic analytics frameworks further enhance situational awareness [31]. Blockchain and federated learning have also been explored to improve security and privacy in smart parking environments [32], [33].

Emerging studies in smart parking have incorporated privacy-preserving learning, real-time vision-based allocation, and reservation-aware management into intelligent parking frameworks [34], [35], [36]. Federated transformer-based slot detection, YOLOv8–OCR vehicle recognition, and competition-reservation coordination have further strengthened perception, operational automation, and service-level parking optimization, respectively.

Despite this extensive body of research spanning early IoT-based architectures, optimization-driven allocation models, medical-facility strategies, and AI-enabled intelligent transportation systems, limited attention has been devoted to grouping-based optimization mechanisms tailored to medical cities. In such environments, allocation decisions must account for heterogeneous vehicle occupancies and the need for balanced parking-space utilization. Existing solutions predominantly optimize either individual vehicle allocation or

economic efficiency, leaving a gap in structured grouping strategies that simultaneously improve utilization efficiency, mitigate congestion, and ensure equitable load distribution. The proposed framework addresses this gap through a grouping-based optimization strategy tailored to medical city operations.

Table 1 summarizes the main contributions of the reviewed literature, presents the key differences among the studies, and highlights how these aspects are addressed in the proposed approach. All reviewed approaches in Table 1 adopted vehicle-based allocation strategies; therefore, this criterion is not reported separately in the comparison.

Table 2 summarizes the main objectives, techniques, evaluation metrics, application domains, and research gaps of existing smart parking approaches and positions the proposed framework within the literature.

A. CRITICAL ANALYSIS AND POSITIONING

Prior smart parking research generally falls into three major streams: (i) infrastructure-centric IoT architectures, (ii) learning-based allocation strategies, and (iii) optimization-based decision models. Although these streams have significantly advanced sensing, monitoring, and routing mechanisms, they rarely incorporate the operational constraints of appointment-driven medical cities, where accessibility, pedestrian density, and crowd distribution are critical.

Infrastructure- and reservation-centric systems: Early IoT-enabled systems emphasized occupancy detection, reservation platforms, and guidance interfaces [7], [10], [11], [17], [20], [21], [37]. However, many of these approaches reported limited quantitative optimization of parking allocation or treated allocation as secondary to monitoring and user interaction. Availability guarantees upon arrival and structured zone balancing were often not explicitly enforced [7]. Consequently, these systems did not systematically mitigate inter-zone imbalances during peak clinical hours, potentially increasing congestion and concentrating pedestrian loads.

Learning-based approaches: Machine learning and reinforcement learning strategies introduced adaptive allocation and guidance policies [13], [15]. While these methods improve adaptability, they typically require extensive training data and may degrade under distributional shifts, such as emergency surges or abnormal scheduling patterns. Moreover, their optimization objectives were predominantly vehicle-centric (e.g., travel time and reward maximization) and did not explicitly incorporate constraints related to medical accessibility or pedestrian crowd distribution.

Optimization-based models: Mathematical and constraint-based formulations improved allocation quality under defined parameters [16], [18]. However, most were designed for generic urban environments and focused on utilization efficiency or economic objectives. Hospital-focused models [22] incorporated behavioral and time-window factors but primarily emphasized profit maximization rather than equitable accessibility.

TABLE 1. Summary of Main Contributions and Research Gaps in Smart Parking Systems

Reference	Main Contribution	Research Gap
[11]	Original research article: smart parking using multi-agent approach, architecture, and workflow	IoT-enabled smart parking system to address urban parking inefficiencies; absence of quantitative evaluation
[13]	An efficient parking management system for smart cities using a GA-RBF model to optimize fuel, time, and cost	Deficient in establishing a comprehensive pipeline integrating data preparation, feature selection, and hybrid model optimization
[7]	A new "smart parking" system infrastructure and implementation	Parking space availability not guaranteed upon drivers' arrival; no optimal allocation based on preferences
[18]	Hierarchical algorithm for in-city parking allocation based on Open Street Map and AnyLogic software	Fragmented addressing of parking allocation and insufficient congestion balancing across networks
[19]	Multi-agent smart parking system with dynamic pricing	Only addresses on-street parking; simplified demand modeling overlooking special events and driver heterogeneity
[20]	IoT-based smart parking system for Indian smart cities	Difficult to judge effectiveness due to lack of quantitative performance evaluation
[15]	Deep reinforcement learning approach toward smart parking architecture	No discussion of deployment-scale constraints, system robustness, and operational costs
[22]	Optimal allocation of shared parking spaces for hospital parkers considering bounded rationality	Focuses only on maximizing profit under time-window constraints; scalability and computational intensity concerns

TABLE 2. Unified Comparison of Smart Parking Research Approaches

Reference	Objective	Techniques	Metrics	Domain	Identified Gap
[7]	Reservation and availability detection	Sensor-based detection with a centralized reservation system	Reservation success rate; availability tracking	Urban	No guaranteed optimal allocation; preference-aware balancing not enforced
[11]	Reduce parking search time and congestion	Multi-agent IoT-based architecture	Qualitative workflow evaluation	Urban	Limited quantitative allocation performance analysis
[13]	Minimize fuel, travel time, and cost	Hybrid GA-RBF optimization model	Fuel usage; travel time; cost	Urban	Lacks structured multi-constraint grouping strategy
[18]	Balance congestion across city parking	Hierarchical multi-criteria allocation using OSM and simulations	Congestion reduction; traffic flow metrics	Urban	Fragmented zone balancing; no passenger-based fairness
[19]	Demand regulation via dynamic pricing	Agent-based dynamic pricing model	Revenue per slot; occupancy rate	Urban	Focuses on economic efficiency rather than equitable accessibility
[20]	IoT-enabled smart parking monitoring	Distributed sensors; cloud processing; mobile app	Detection accuracy; response time	Urban	Monitoring-centric; allocation treated as secondary component
[15]	Adaptive parking allocation via reinforcement learning	Deep Q-Network (DQN) simulation-based policy learning	Cumulative reward; convergence time	Urban	Vehicle-centric optimization; limited domain-specific constraints
[22]	Maximize hospital parking profit under time windows	Cumulative Prospect Theory allocation model	Profit; adherence to time constraints	Medical	Revenue-focused; does not prioritize equitable people-based load distribution
[5]	Efficiency–equity trade-off in reservations	Mathematical reservation modeling	Equity index; waiting time; utilization rate	Urban	Equity analyzed abstractly; not operationalized via passenger-based load metric
[16]	Real-time constraint-based allocation	Mathematical constraint optimization model	Allocation efficiency	Urban	Generic urban setting; lacks medical-specific operational constraints
Proposed Work	Enhance utilization and accessibility in medical cities	Grouping-based load-balancing optimization (six heuristic algorithms) with IoT integration	Best-solution percentage (P_c); average load gap (A_g); execution time (< 0.001 s)	Medical cities	Introduces a passenger-based fairness metric and a medical-city-specific grouping strategy

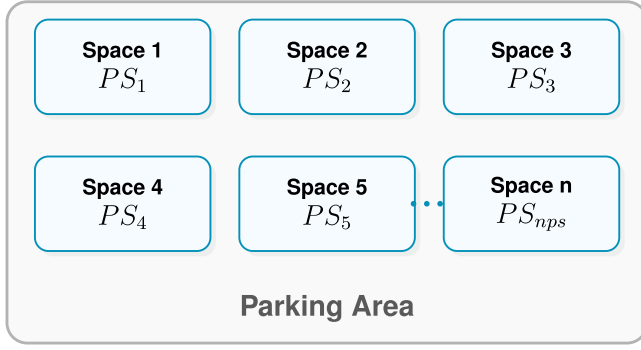


FIGURE 1. Structure of the parking area divided into parking spaces.

Equity and load-balancing perspective: Equity-aware reservation models analyzed efficiency–equity trade-offs [5], and load-balancing principles have been extensively studied in networking and computing domains [23], [24], [25]. Yet parking allocation studies have rarely operationalized fairness through a people-based load metric that captures downstream pedestrian concentration. This limitation is particularly critical in medical cities, where passenger count, escort presence, and patient condition directly influence accessibility and crowd dynamics.

Positioning of this work: In contrast to prior approaches, this paper formulates allocation fairness through passenger-count-based zone balancing. By minimizing inter-zone disparity in the total number of people rather than vehicles, the proposed framework explicitly links parking allocation to accessibility and crowd distribution. The four-stage grouping procedure and six grouping–permutation heuristics provide computational efficiency, with sub-millisecond execution times, making the approach suitable for real-time deployment in medical cities. Security- and privacy-oriented enhancements (e.g., blockchain and federated learning [32], [33]) are complementary but orthogonal to the core contribution, which lies in the people-centric objective and lightweight optimization design.

III. PROBLEM DESCRIPTION

The proposed parking area is divided into multiple parking spaces, each denoted by PS_i , where $i \in \{1, 2, \dots, n_{ps}\}$ and n_{ps} is the total number of available parking spaces, as illustrated in Fig. 1. These spaces may be isolated or distributed across different zones within the medical city.

This modular configuration supports efficient management and flexible use of parking resources. The parking spaces need not be physically contiguous; instead, they may be strategically distributed across separate zones within the medical city. At the operational level, allocation decisions are made at the parking-space level, whereas parking lots serve only to describe the higher-level physical organization of the parking area.

Fig. 2 illustrates the structural organization of the designated parking facility. The parking area comprises multiple

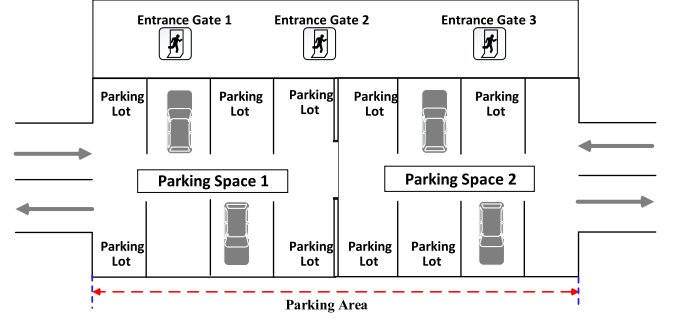


FIGURE 2. Illustration of Parking Space and Lot Organization Within a Parking Area.

parking lots, each containing several parking spaces for vehicle accommodation. Vehicles are assigned to available parking spaces based on space availability and the proposed load-balancing strategy.

Multiple entry gates regulate vehicle access to maintain smooth traffic flow and reduce congestion. Each parking space is uniquely identified, enabling structured allocation and real-time occupancy monitoring. This modular layout supports dynamic parking assignment, improves space utilization, and reduces visitor search time, making it particularly well suited to high-demand environments such as medical cities.

Remark 1: This manuscript focuses on the assignment of vehicles to available parking spaces across diverse environments rather than on directing vehicles to fixed or predesignated parking facilities.

This study allocates vehicles to parking spaces according to the total number of people associated with each assignment rather than according to the number of vehicles alone.

Let:

- Ve_j denotes vehicle j ,
- n_v denotes the total number of vehicles,
- np_j denotes the number of people inside vehicle Ve_j ,
- PSL_i denotes the total number of people assigned to the parking space PS_i (parking space load).

The parking space load is defined as $PSL_i = \sum_{j=1}^{n_v} np_j y_{ij}$, $\forall i \in \{1, \dots, n_{ps}\}$.

The primary objective is to minimize the disparity between parking space loads. The overall discrepancy across all parking spaces is defined as:

$$Tg = \sum_{i=1}^{n_{ps}} [PSL_i - PSL_{min}] \quad (1)$$

The aim is to determine a schedule that minimizes Tg . The binary decision variable is defined as:

$$y_{ij} = \begin{cases} 1 & \text{if vehicle } j \text{ is assigned to parking space } i, \\ 0 & \text{otherwise.} \end{cases} \quad (2)$$

The optimization model is therefore formulated as follows:

$$\min Tg \quad (3)$$

subject to

$$\sum_{i=1}^{n_{ps}} y_{ij} = 1, \forall j \in \{1, \dots, n_v\} \quad (4)$$

$$PSL_i \geq PSL_{min}, \forall i \in \{1, \dots, n_{ps}\} \quad (5)$$

$$y_{ij} \in \{0, 1\}, \forall i \in \{1, \dots, n_{ps}\}, \forall j \in \{1, \dots, n_v\} \quad (6)$$

$$PSL_{min} \geq 0 \quad (7)$$

Constraint (4) ensures each vehicle is assigned to exactly one parking space. Constraint (5) defines the minimum load condition for each parking space. Constraint (6) enforces binary assignment. Constraint (7) ensures non-negativity.

IV. PROPOSED SYSTEM

Technology in medical cities is employed to enhance health-care accessibility, reduce patient burden, and improve overall service quality. The intelligent parking allocation framework integrates appointment scheduling, real-time sensing, and grouping-based optimization to manage parking resources efficiently under dynamic operational conditions.

Medical city beneficiaries (patients or visitors) can register their parking requests through web- or mobile-based applications. During registration, the system collects the information required for allocation and access validation, including:

- number of passengers;
- transportation mode (private vehicle or other means); and
- vehicle and booking identification data.

After registration, the system generates a QR code containing the booking identifier, beneficiary ID, declared passenger count, and allocated parking space number.

Upon arrival, the beneficiary scans the QR code using gate-level sensors installed at the main entrances to the parking facilities. These devices validate the booking data and compare the declared number of passengers with the actual vehicle occupancy detected by cameras and embedded sensors. After validation, the following actions are performed:

- a guidance card is printed;
- a copy is transmitted to the scheduler;
- the allocation status is forwarded to the monitoring subsystem; and
- guidance information is displayed on digital panels.

The scheduler continuously updates the status of parking spaces using real-time data from the control unit, including:

- gate cameras;
- in-space occupancy sensors;
- mobile applications; and
- medical city information systems.

Visitors who arrive without vehicles are directed to their destinations without affecting parking-space load status.

In contrast to conventional systems that rely on vehicle counts, the proposed allocation strategy assigns vehicles on the basis of passenger count, thereby promoting a more balanced distribution of people across parking areas. This

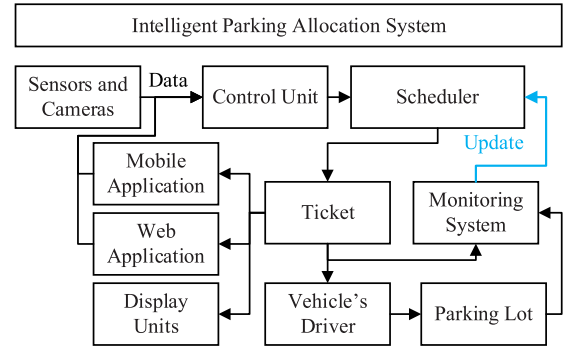


FIGURE 3. Architecture of the Intelligent Parking Allocation System.

approach helps mitigate pedestrian crowding after vehicle disembarkation and supports smoother internal mobility within the medical city.

The equitable distribution of individuals across parking spaces:

- reduces congestion at entry points;
- minimizes walking delays to clinics;
- prevents interference with emergency cases; and
- supports coordinated internal crowd management.

Fig. 3 presents the block diagram of the intelligent parking allocation system.

A. COMPONENT INTERACTION AND DATA FLOW

The system follows a structured multi-layer interaction model composed of five primary components:

- 1) Booking Interface
- 2) Scheduler
- 3) Allocation Engine
- 4) Control Unit
- 5) Monitoring and Guidance Subsystem

Step 1-Data Submission Vehicle-related information, including passenger count np_j , is submitted through the booking interface and transmitted to the scheduler:

Step 2-Allocation Processing: The scheduler forwards validated requests to the allocation engine, which executes the grouping-based optimization model (Equations 1–7). The algorithm assigns vehicle Ve_j to parking space PS_i to minimize load imbalance Tg . The assigned space is then encoded in the QR code.

Step 3-Entry Validation: At the entry gates, QR-code scanning and occupancy sensing are used to validate real-time passenger data. The control unit verifies the consistency between scheduled and observed values.

Step 4-Status Synchronization: Parking-space loads PSL_i are updated in real time. Any discrepancy triggers corrective actions, such as reallocation or monitoring flags.

Step 5-Monitoring and Guidance: Allocation decisions and occupancy status are propagated to digital signage systems. The monitoring subsystem ensures closed-loop synchronization among booking data, optimization decisions, and physical parking operations. This architecture guarantees

consistency among real-time sensing, scheduling decisions, and optimization objectives.

B. OPERATIONAL ADAPTABILITY

The proposed framework supports dynamic parking allocation in medical city environments through real-time status updates and lightweight heuristic decision-making. Its very low execution time suggests potential applicability in high-demand settings. However, robustness under surge-arrival conditions was not explicitly evaluated in this study and therefore remains an important direction for future research.

In real-world medical city operations, highly uneven vehicle arrival patterns may arise during peak clinical hours or emergency situations. Given its grouping-based structure and low computational complexity, the proposed framework may remain responsive under such conditions. Nevertheless, this assumption was not examined experimentally in the present study.

V. NOVEL GROUPING ROUTINE

The proposed routine follows a sequential process: the initial arrangement of vehicles is defined, vehicles are classified into three groups (G_1 , G_2 , and G_3), a permutation procedure generates the scheduling sequence, and vehicles are assigned to parking spaces according to this sequence. The detailed steps are as follows.

- *Step 1:* Define the initial spatial arrangement of the vehicles.
- *Step 2:* Organize and classify the vehicles into three distinct groups, denoted G_1 , G_2 , and G_3 , to support the subsequent stages of the routine.
- *Step 3:* Generate a scheduling sequence based on the parameters established in the preceding steps.
- *Step 4:* Assign vehicles to parking spaces according to the generated scheduling sequence to optimize space utilization.

The following subsections provide detailed descriptions of the first three steps of the proposed routine.

A. SPECIFYING THE INITIAL ARRANGEMENT OF VEHICLES

The initial arrangement of vehicles is critical to group formation in the proposed routine. Three distinct methods are introduced to define the initial vehicle configuration, as outlined below.

- *Way 1 (Arrival Sequence):* Vehicles are scheduled in their original arrival order without any reordering.
- *Way 2 (Ascending Occupancy):* Vehicles are sorted in ascending order based on their passenger count (np_j).
- *Way 3 (Descending Occupancy):* Vehicles are ordered such that $np_1 \geq np_2 \geq \dots \geq np_n$.

B. CREATION OF THE THREE GROUPS G_1 , G_2 , AND G_3

The novel routine is based on the grouping method. Vehicles are divided into three groups, denoted G_1 , G_2 , and G_3 , using a specific method. Initially, these groups are empty. The numbers of vehicles in G_1 , G_2 , and G_3 are denoted by nv_1 , nv_2 ,

and nv_3 , respectively. In practice, $nv_1 = \frac{nv}{3}$, $nv_2 = \frac{nv}{3}$, and $nv_3 = nv - nv_1 - nv_2$. Consequently,

$$G_1 = \{Ve_1, \dots, Ve_{nv_1}\},$$

$$G_2 = \{Ve_{nv_1+1}, \dots, Ve_{nv_1+nv_2}\},$$

$$G_3 = \{Ve_{nv_1+nv_2+1}, \dots, Ve_{nv}\}.$$

The composition of G_1 , G_2 , and G_3 is determined by the initial arrangement of the vehicles.

For clarity, the group construction is summarized as follows: G_1 consists of the first nv_1 vehicles, G_2 consists of the next nv_2 vehicles, and G_3 consists of the remaining nv_3 vehicles. Accordingly, the three groups are defined once the initial vehicle arrangement is specified.

C. CHOOSING THE PERMUTATION TO GENERATE THE SCHEDULING SEQUENCE

For each unique vehicle configuration, three distinct groups (G_1 , G_2 , and G_3) are formed. Subsequently, all possible permutations of the sequence defined by the tuple (G_1, G_2, G_3) are generated. There are six possible permutations for arranging these groups. The first configuration, denoted G_{123} , schedules all vehicles from G_1 first, followed by G_2 , and then G_3 .

The second permutation is G_{132} , which schedules the groups in the order G_1 , G_3 , and G_2 . In this sequence, vehicles in G_1 are scheduled first, followed by those in G_3 , and finally those in G_2 . The third permutation is G_{213} , which schedules the groups in the order G_2 , G_1 , and G_3 . Thus, vehicles in G_2 are scheduled first, followed by those in G_1 , and then those in G_3 . The fourth permutation is G_{231} , which schedules the groups in the order G_2 , G_3 , and G_1 . The fifth permutation is G_{312} , which schedules the groups in the order G_3 , G_1 , and G_2 . The sixth permutation is G_{321} , which schedules the groups in the order G_3 , G_2 , and G_1 .

VI. PROPOSED ALGORITHMS

This section presents the proposed algorithms for parking space allocation in medical city environments. The proposed algorithms classify vehicles into groups and schedule them according to different permutation strategies. Each algorithm follows a standardized structure, beginning with the initial configuration of vehicles, followed by group formation, permutation selection, and vehicle scheduling. The main objective of these algorithms is to reduce load disparities among parking spaces based on vehicle occupancy. Details of the algorithms and their pseudocode are provided in the following subsections.

The primary objective of these algorithms is to minimize load disparities among parking spaces by considering vehicle occupancy.

Based on the proposed framework, six algorithms are developed. For every conceivable initial configuration of the vehicles, a corresponding permutation from one of the six potential groupings is systematically applied to ensure the best arrangement and organization. In the following context, the

TABLE 3. Permutation Used in Each Proposed Algorithm

Permutation: G_{xyz}	Algorithm
$G123$	GVF
$G132$	GV_S
$G213$	GVT
$G231$	GVO
$G312$	GVI
$G321$	GVX

Algorithm 1: Grouping Vehicles Using the $G123$ Permutation: Algorithm GVF .

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1: for ( $way = 1$  to  $3$ ) do
2:   if ( $way = 2$ ) then
3:     Call  $IRS(ve)$ 
4:   end if
5:   if ( $way = 3$ ) then
6:     Call  $DRS(ve)$ 
7:   end if
8:   Call  $GS(ve)$ 
9:   Call  $SLH(G123)$ 
10:  Calculate  $Tg_{way}$ 
11: end for
12: Calculate  $Tg = \min_{1 \leq way \leq 3}(Tg_{way})$ 
13: Return  $Tg$ 

```

computational method used to sort a designated list, denoted by the variable L , in descending order based on its values will be officially denoted as $DRS(L)$, indicating the process of organizing the list's elements from the peak to the bottom.

Similarly, the computational procedure that is utilized to organize the same list L in ascending order, wherein the values are arranged from the lowest to the highest, will be appropriately denoted as $IRS(L)$, thus establishing a clear distinction between the two sorting methodologies based on their respective ordering criteria. The procedure that systematically partitions an input list L into three distinct groups, thereby facilitating further analysis or processing, is conventionally denoted $GS(L)$ and encapsulates the essence of grouping in list manipulation. Finally, the procedure that schedules the aforementioned three groups under a specified permutation represented by the variable G_{xyz} will be denoted as $SLH(G_{xyz})$, thereby indicating the importance of the order in which these groups are arranged according to the defined permutation structure.

Table 3 presents the permutation used in each proposed algorithm.

As shown in Table 3, the algorithm grouping vehicles using permutation $G123$ is GVF . This algorithm identifies and selects the specific permutation $G123$. This algorithm, denoted as GVF (Grouping Vehicles Using the $G123$ Permutation), is detailed in Algorithm 1.

As shown in Table 3, the algorithm grouping vehicles using permutation $G132$ is GV_S . The third algorithm selects

permutation $G213$ and is denoted by GVT . The algorithm GVT follows the same structure as Algorithm 1, except that Step 9 is changed from “Call $SLH(G132)$ ” to “Call $SLH(G213)$ ”.

The fourth algorithm is characterized by its selection of the permutation denoted as $G231$. This algorithm is formally referred to as GVO . The fifth algorithm uses the permutation as $G312$. This algorithm is formally referred to in the literature as GVI . The algorithm identified as the sixth in the series of computational procedures selects the permutation as $G321$. The process in this algorithm was formally referred to as GVX . The complexity of the proposed algorithm is $O(n \log n)$.

VII. EXPERIMENTAL RESULTS

A. MATERIALS AND INSTANCE GENERATION

The proposed algorithms were implemented in Microsoft Visual C++ and executed on a workstation equipped with an Intel(R) Core i5 CPU running at 3.40 GHz and 8 GB of RAM. To enable a controlled and reproducible evaluation, we followed the benchmark instance-generation protocol adopted in [6]. For each instance, n_v vehicles were generated, and each vehicle j was assigned a passenger count np_j .

We tested the following problem scales: $n_v \in \{10, 25\}$ with $np_s \in \{3, 4, 5\}$; $n_v \in \{50, 70, 90, 110\}$ with $np_s \in \{5, 15, 25, 35\}$; and $n_v \in \{200, 400, 600\}$ with $np_s \in \{20, 40, 60\}$. For each tuple $(n_v, np_s, \text{Category})$, 10 independent instances were generated, yielding a total of $(2 \times 3 + 4 \times 4 + 3 \times 3) \times 10 \times 9 = 2,790$ instances. Although a live field deployment was outside the scope of this study, these benchmark categories provided diverse demand profiles that stress-tested the allocation procedure under different occupancy mixes and problem scales.

B. METRICS

To assess solution quality and efficiency, we used the following metrics:

- V^* : The best value obtained among all algorithms after executing all of them.
- V : The output generated by the algorithm currently under evaluation.
- P_c : The percentage of instances for which $V = V^*$.
- $gp = \frac{V - V^*}{V}$, with $gp = 0$ when $V = 0$.
- Ag : The mean value of the gap gp computed across a specified number of instances.
- $Time$: The execution time measured in seconds, or “+” if the execution time is less than 0.001 seconds.

C. COMPARISON OF THE PROPOSED HEURISTICS

The overall results are summarized in Table 4, and the sensitivity analyses with respect to n_v , np_s , and instance categories are reported in Tables 5–8.

As illustrated in Table 4, the algorithm GVF outperforms the alternative algorithms across all evaluation metrics. It achieves the highest best-solution percentage ($P_c = 68.2\%$)

TABLE 4. Performance Comparison of the Proposed Algorithms Based on Best Solution Rate, Average Gap, and Execution Time

	<i>GVF</i>	<i>GVS</i>	<i>GVT</i>	<i>GVO</i>	<i>GVI</i>	<i>GVX</i>
P_c	68.2%	21.0%	43.5%	16.5%	20.0%	20.1%
Ag	0.137	0.452	0.289	0.523	0.452	0.464
<i>Time</i>	+	+	+	+	+	+

TABLE 5. Variation of Average Gap (Ag) With Respect to the Number of Vehicles (n_v) for All Algorithms

n_v	<i>GVF</i>	<i>GVS</i>	<i>GVT</i>	<i>GVO</i>	<i>GVI</i>	<i>GVX</i>
10	0.153	0.310	0.250	0.377	0.247	0.250
25	0.268	0.539	0.341	0.532	0.426	0.465
50	0.035	0.390	0.201	0.436	0.424	0.380
70	0.137	0.512	0.249	0.543	0.367	0.448
90	0.217	0.250	0.302	0.403	0.317	0.290
110	0.122	0.520	0.348	0.597	0.597	0.588
200	0.094	0.594	0.315	0.678	0.617	0.642
400	0.073	0.693	0.359	0.727	0.746	0.748
600	0.150	0.309	0.254	0.455	0.361	0.421

TABLE 6. Variation of Average Gap (Ag) With Respect to the Number of Parking Spaces (n_{ps}) for All Algorithms

n_{ps}	<i>GVF</i>	<i>GVS</i>	<i>GVT</i>	<i>GVO</i>	<i>GVI</i>	<i>GVX</i>
3	0.276	0.465	0.335	0.432	0.371	0.363
4	0.262	0.339	0.344	0.391	0.273	0.256
5	0.126	0.586	0.295	0.666	0.566	0.612
15	0.233	0.425	0.300	0.563	0.376	0.473
20	0.078	0.570	0.304	0.631	0.573	0.625
25	0.048	0.332	0.291	0.402	0.386	0.334
35	0.086	0.270	0.170	0.286	0.276	0.206
40	0.172	0.569	0.324	0.665	0.570	0.617
60	0.067	0.457	0.299	0.564	0.582	0.570

TABLE 7. Variation of Average Gap (Ag) Across Different Instance Classes for All Algorithms

Category	<i>GVF</i>	<i>GVS</i>	<i>GVT</i>	<i>GVO</i>	<i>GVI</i>	<i>GVX</i>
1	0.102	0.338	0.365	0.596	0.456	0.492
2	0.004	0.413	0.195	0.671	0.491	0.514
3	0.011	0.379	0.262	0.666	0.475	0.512
4	0.269	0.423	0.225	0.414	0.405	0.414
5	0.293	0.547	0.219	0.526	0.493	0.479
6	0.251	0.557	0.226	0.530	0.472	0.471
7	0.080	0.441	0.332	0.412	0.401	0.406
8	0.103	0.482	0.359	0.445	0.428	0.442
9	0.122	0.491	0.416	0.451	0.448	0.451

and the lowest average gap ($Ag = 0.137$), thereby demonstrating superior effectiveness in optimizing parking-space load balancing.

The second-best algorithm is *GVT*, with $P_c = 43.5\%$ and $Ag = 0.289$, although it remains inferior to *GVF*. The other algorithms (*GVS*, *GVO*, *GVI*, and *GVX*) achieved lower P_c values and larger average gaps, indicating weaker performance. In terms of computational efficiency, all algorithms required less than 0.001 seconds, confirming their suitability for real-time parking allocation.

As shown in Table 5, the algorithm *GVF* yields the lowest average gap (Ag) across all evaluated values of n_v , thereby

TABLE 8. Variation of Average Gap for All Proposed Algorithms Across Different Numbers of Vehicles (n_v) and Parking Spaces (n_{ps})

<i>PA</i>	n_v	n_{ps}	<i>GVF</i>	<i>GVS</i>	<i>GVT</i>	<i>GVO</i>	<i>GVI</i>	<i>GVX</i>
1		3	0.302	0.328	0.458	0.405	0.251	0.250
2	10	4	0.127	0.333	0.174	0.312	0.280	0.239
3		5	0.030	0.268	0.117	0.414	0.210	0.260
4		3	0.250	0.602	0.212	0.459	0.492	0.476
5	25	4	0.396	0.344	0.514	0.471	0.266	0.273
6		5	0.158	0.669	0.296	0.667	0.521	0.644
7		5	0.050	0.712	0.318	0.793	0.816	0.796
8	50	15	0.080	0.200	0.292	0.396	0.313	0.302
9		25	0.010	0.377	0.194	0.491	0.296	0.354
10		35	0.000	0.271	0.000	0.066	0.271	0.066
11		5	0.140	0.863	0.357	0.845	0.679	0.805
12	70	15	0.293	0.599	0.244	0.617	0.225	0.559
13		25	0.104	0.203	0.188	0.201	0.247	0.068
14		35	0.010	0.383	0.207	0.510	0.317	0.360
15		5	0.321	0.184	0.363	0.405	0.284	0.284
16	90	15	0.373	0.310	0.451	0.593	0.297	0.382
17		25	0.069	0.180	0.287	0.325	0.354	0.297
18		35	0.103	0.325	0.106	0.289	0.332	0.195
19		5	0.059	0.820	0.317	0.871	0.889	0.881
20	110	15	0.186	0.590	0.213	0.645	0.667	0.648
21		25	0.009	0.569	0.494	0.592	0.647	0.619
22		35	0.233	0.102	0.368	0.280	0.184	0.204
23		20	0.080	0.791	0.357	0.845	0.871	0.849
24	200	40	0.186	0.769	0.317	0.765	0.635	0.718
25		60	0.017	0.222	0.272	0.423	0.344	0.360
26		20	0.044	0.909	0.445	0.893	0.827	0.878
27	400	40	0.066	0.805	0.375	0.859	0.881	0.858
28		60	0.109	0.364	0.257	0.430	0.531	0.509
29		20	0.111	0.010	0.111	0.154	0.020	0.148
30	600	40	0.263	0.132	0.281	0.372	0.193	0.276
31		60	0.074	0.786	0.369	0.838	0.871	0.841

demonstrating its superior performance and scalability. The Ag values of *GVF* remain low even as the number of vehicles increases.

Algorithm *GVT* is moderately competitive, maintaining the second-lowest Ag values across different vehicle loads. In contrast, *GVS*, *GVO*, *GVI*, and *GVX* yield higher gap values, particularly as n_v increases, indicating weaker load-balancing performance. In particular, under conditions of high traffic intensity ($n_v = 400$ and $n_v = 600$), the performance of *GVO*, *GVI*, and *GVX* declines markedly, further confirming that *GVF* is the most reliable and effective algorithm for achieving equitable parking-space allocation across varying traffic scenarios.

Fig. 4 illustrates the variation in the average gap (Ag) with respect to the number of vehicles (n_v) for all proposed algorithms. It can be observed that *GVF* consistently achieves the lowest gap values across all tested levels of n_v . In contrast, *GVS*, *GVO*, and *GVX* exhibit larger gaps, particularly at higher values of n_v . Overall, *GVF* demonstrates superior scalability and robustness compared with the other algorithms.

As shown in Table 6, the algorithm *GVF* yields the lowest mean gap (Ag) across all values of n_{ps} , demonstrating a strong capacity to adapt to varying numbers of parking spaces.

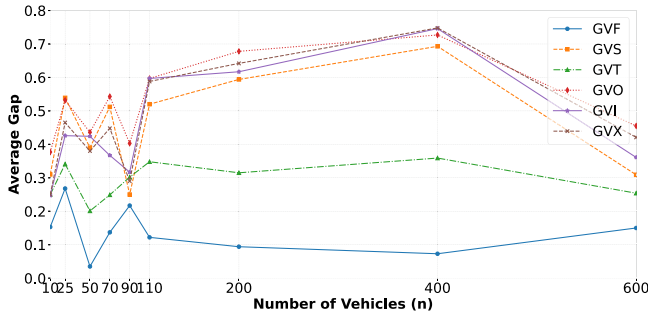


FIGURE 4. Variation of Average Gap (Ag) with Respect to the Number of Vehicles (n).

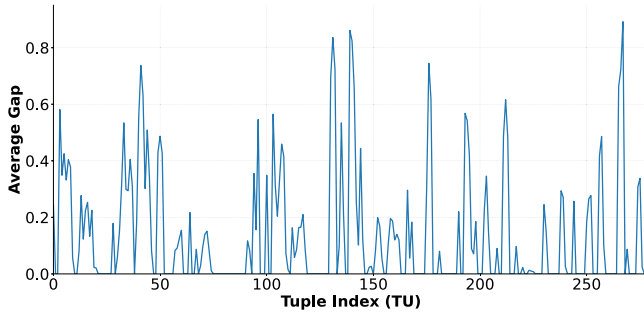


FIGURE 5. Average Gap Variation of *GVF* Across Different Tuples (n_v, n_{ps} , Category).

Its performance remains stable and improves as the number of parking spaces increases, with particularly low Ag values observed at $n_{ps} = 25$ and $n_{ps} = 60$. In contrast, although *GVT* is moderately competitive, it consistently remains inferior to *GVF*. The other algorithms (*GVS*, *GVO*, *GVI*, and *GVX*) exhibit substantially higher gap values, especially at both lower and higher values of n_{ps} , indicating weaker load-balancing performance. These findings suggest that *GVF* provides the most reliable and scalable solution for the equitable allocation of parking spaces across diverse parking configurations.

As shown in Table 7, the algorithm *GVF* attains the lowest average gap (Ag) across most instance categories, thereby demonstrating remarkable robustness and adaptability. For classes 2 and 3, *GVF* achieves near-optimal performance, with Ag values approaching zero, whereas all alternative algorithms exhibit larger gap values. Although the Ag values of *GVF* increase slightly in classes 4 to 6, it still outperforms all other methods by a significant margin. In classes 7 to 9, *GVF* maintains superior performance relative to the competing algorithms, which exhibit larger and less consistent gap values. These results indicate that *GVF* is the most reliable algorithm across the tested scenarios, whereas *GVT* is comparatively less competitive.

Fig. 5 illustrates the variation in the mean gap values achieved by the algorithm *GVF* across different tuples

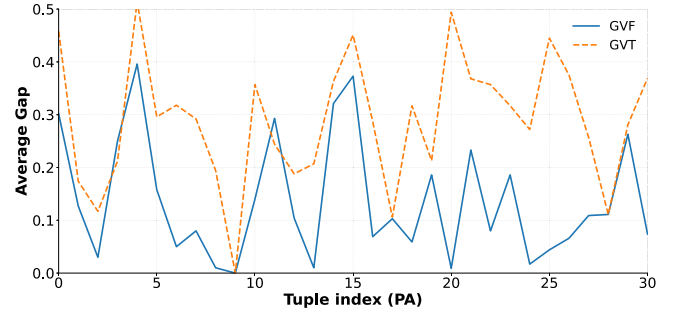


FIGURE 6. Comparison of Average Gap Values Between *GVF* and *GVT* Across Different PA Values.

TU , where each TU represents a specific combination of (n_v, n_{ps} , Category). A total of 279 tuples were evaluated. The results show that, for most tuples, the algorithm achieves an average gap of 0.00, indicating highly effective load distribution across parking spaces. Notably, 139 of the 279 tuples yielded a gap of 0.00, representing the most favorable conditions. In contrast, only a limited subset of tuples exhibited the highest gap values: 14 distinct TU instances showed considerably larger gaps, corresponding to the least favorable conditions. Overall, Fig. 5 confirms the robustness of *GVF* in achieving equitable parking allocation across diverse configurations, with only a small number of outlier scenarios showing increased gap values.

Fig. 6 compares the mean gap values obtained by *GVF* and *GVT* across the tuple indices PA , where each PA represents a specific combination of (n_v, n_{ps}). In total, 31 PA values were considered. The findings show that *GVF* consistently outperforms *GVT*, achieving lower mean gap values across most PA instances. Although both algorithms exhibit performance variations, *GVF* demonstrates more consistent low-gap behavior, whereas *GVT* exhibits recurrent higher peaks, indicating greater variability and weaker load-balancing performance. The algorithm *GVF* also achieves near-zero gap values at several PA indices, further highlighting its robustness. This comparative analysis demonstrates the superior efficiency, scalability, and adaptability of *GVF* in managing diverse parking allocation scenarios.

For the tuple index PA , both algorithms, *GVF* and *GVT*, attain a gap value of 0.00 only at $PA = 10$. Moreover, the gap values of *GVF* remain lower than those of *GVT* for all other PA values. In addition, *GVF* more frequently produces gap values close to zero, indicating better load-balancing performance across diverse parking allocation settings.

Table 8 demonstrates that the algorithm *GVF* achieves the lowest average gap (Ag) values across a wide range of vehicle and parking-space configurations, indicating robust performance for both small- and large-scale instances. In contrast, *GVS*, *GVO*, and *GVX* frequently exhibit larger gap values as the vehicle count increases, reflecting weaker load-balancing performance. Although *GVT* is moderately competitive, it remains less effective than *GVF*. These results confirm that

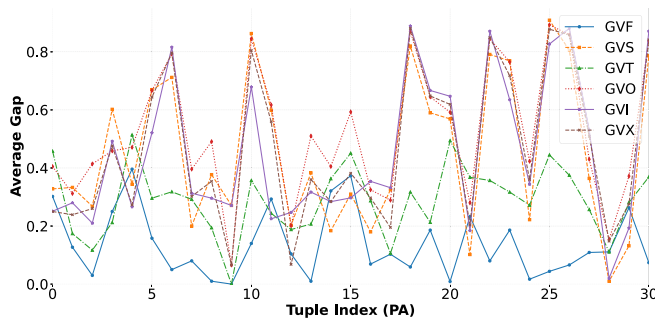


FIGURE 7. Variation of the average gap values for all proposed algorithms across different tuple indices (PA).

GVF is the most robust and scalable method for dynamic and equitable parking allocation in complex medical city environments.

Fig. 7 illustrates the variation of the average gap values for all proposed algorithms (*GVF* to *GVX*) across different tuple indices (PA).

D. COMPARISON WITH STATE-OF-THE-ART APPROACHES

To provide a clear comparison with state-of-the-art methods, we conducted a direct head-to-head evaluation against the literature baseline algorithm *MR*, which was designed for the same objective and benchmark setting [6].

Let GV_b denote the best-performing variant among the proposed heuristics.

Across the 2,790 benchmark instances, the results show that the best proposed heuristic variant, GV_b , matched or outperformed the baseline *MR* in 1,547 instances, representing 55.4% of the test set. More specifically, GV_b outperformed *MR* in 57 instances (2.0%) and matched it in 1,490 instances (53.4%), whereas *MR* performed better in 1,243 instances (44.6%). These results indicate that the proposed heuristic family was competitive with the baseline in more than half of the instances while requiring substantially lower execution times, thereby supporting its suitability for real-time parking allocation.

VIII. CONCLUSION

This paper introduced a people-centric intelligent parking allocation framework tailored specifically to medical city environments. In contrast to conventional vehicle-based allocation strategies, the proposed approach formulates parking assignment as a load-balancing problem based on passenger count, thereby directly linking parking decisions to accessibility and internal pedestrian-flow management. This perspective enables a more equitable distribution of visitors across parking zones and mitigates post-parking congestion near high-demand clinics.

A four-step grouping routine was developed to implement the proposed allocation model. The routine consists of defining the initial vehicle arrangement, partitioning vehicles into three groups, selecting a group permutation, and

assigning vehicles to parking spaces. Based on this routine, six grouping-permutation heuristic algorithms were designed and evaluated on 2,790 benchmark instances across nine instance classes. Among them, *GVF* achieved the best overall performance, producing the best solutions in 68.2% of the instances, with an average gap of 0.137 and execution times below 0.001 s.

The computational results demonstrated the effectiveness of the framework across the considered benchmark instance classes, with stable performance and sub-millisecond execution times. The proposed framework integrates IoT sensing, real-time validation, and lightweight optimization to support operational feasibility in dynamic healthcare settings. By aligning parking allocation with patient accessibility requirements and crowd distribution, the system improves traffic flow, reduces congestion, and enhances the service experience in medical cities.

Future research will focus on extending the model to multi-objective formulations that incorporate walking distance, appointment urgency, and emergency prioritization. Additional work will investigate theoretical lower bounds for performance benchmarking and develop predictive, demand-aware allocation strategies using machine learning. A real-world pilot deployment incorporating on-site IoT occupancy sensing and live appointment logs is also planned to validate system behavior under real traffic dynamics, operational constraints, and user behavioral variability.

ACKNOWLEDGMENT

The authors would like to acknowledge the support of Prince Sultan University for paying the Article Processing Charges (APC) of this publication.

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